OLS Mechanics I

Derivation, Fit, and Optimality

Gov 2001 · Scott Cunningham · April 15, 2026

Every four years, economists ask whether the economy elects presidents

- **1948:** Economy growing → Truman wins
- **1980:** Recession → Carter loses to Reagan
- **1992:** "It's the economy, stupid" → Clinton wins
- **2008:** Financial crisis → Obama wins

Fair (1978)

"The Effect of Economic Events on Votes for President"

Review of Economics and Statistics

Fair quantified this intuition: 24 elections, 1916–2016. Today we work through exactly how.

Ray Fair spent five decades studying how economic conditions shape presidential elections

- **Ray Fair** — John M. Musser Professor of Economics, Yale University
- **Sharon Oster** (Yale School of Management) — his wife, pioneering industrial economist; died June 10, 2022
- **Emily Oster** (Harvard PhD, 2005) — their daughter; Professor at Brown; author of *Expecting Better*, *Cribsheet*
- **Jesse Shapiro** (Harvard PhD, 2005) — son-in-law; Harvard Professor; MacArthur Fellow, 2021

An economics dynasty

Two Harvard PhDs (2005)

One MacArthur Fellow

One bestselling author

All connected to Yale

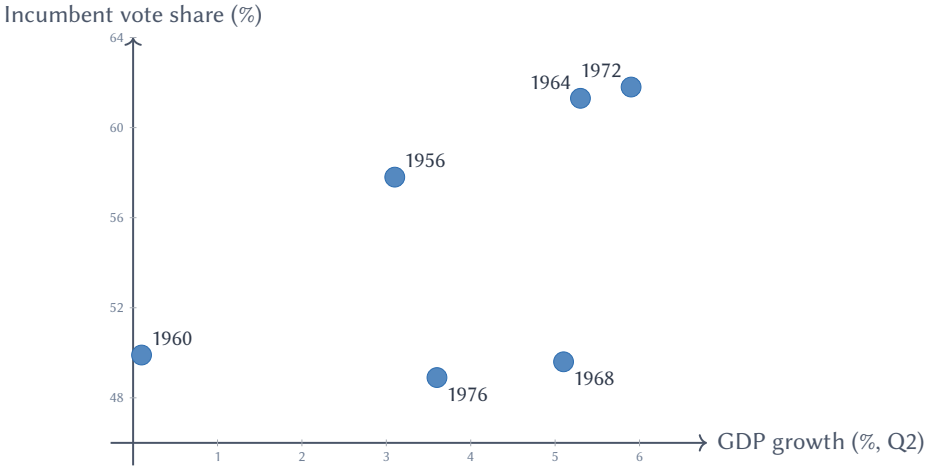
As of 2024, Fair's model still forecasts presidential elections on his Yale website.

Fair's data: six elections, three predictors, one outcome

Year	GDP growth (%)	Inflation (%)	Vote share (%)	Incumbent?
1956	3.1	2.1	57.8	Yes
1960	0.1	2.5	49.9	No
1964	5.3	1.1	61.3	No
1968	5.1	3.5	49.6	No
1972	5.9	4.8	61.8	Yes
1976	3.6	6.6	48.9	Yes

Y = incumbent party presidential vote share (%); GDP growth = Q2 of election year. Full dataset: $n = 24$ elections, 1916–2016.

The pattern: good economies re-elect incumbents



OLS estimates the BLP coefficient β — here is what it finds

Variable	$\hat{\beta}$	SE	Interpretation
Intercept	46.3	1.8	baseline vote (%)
GDP growth	+0.70	0.23	+1% growth $\Rightarrow$ +0.7 pp
Inflation	−0.68	0.22	+1% inflation $\Rightarrow$ −0.7 pp
Incumbent	+4.0	1.4	incumbency bonus: +4 pp

$n = 24$ elections (1916–2016), $R^2 = 0.73$

Estimand: β (BLP in the population) **Estimator:** OLS **Estimate:** $\hat{\beta}$ above

Today: how does OLS produce $\hat{\beta}$ from data?

Running Fair's model in R takes two lines

```r
# Load Fair election data (n=24, 1916-2016)
fair <- read.csv("fair_elections.csv")

# Fit the OLS model
fit <- lm(vote ~ gdpg + inflation + incumbent,
          data = fair)

summary(fit)
```

`lm()` minimizes the sum of squared residuals. Everything else in today's lecture explains *why* that works.

OLS is the BLP's sample analog: replace $\mathbb{E}[\,\cdot\,]$ with $\frac{1}{n}\sum_i$

Population BLP

$$\beta_1 = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}$$

uses $\mathbb{E}[\,\cdot\,]$

$\xrightarrow{\text{DATA}}$

Sample OLS

$$\hat{\beta}_1 = \frac{\hat{S}_{XY}}{\hat{S}_{XX}}$$

uses $\frac{1}{n}\sum_i$

Plug-in principle: same minimization problem, sample moments replace population moments

Today: four building blocks

1. **Derivation** — bivariate FOCs $\longrightarrow \hat{\beta}_1 = \hat{S}_{XY}/\hat{S}_{XX}$
2. **Matrix form** — $\hat{\beta} = (X'X)^{-1}X'Y$
3. **Model fit** — R^2, adjusted R^2, overfitting
4. **Gauss–Markov** — when OLS is BLUE

Part I: Deriving $\hat{\beta}$

Minimize squared residuals; solve two equations

Notation: (α, β_1) are the true parameters; (a, b) are the candidate values we search over

The model:

$$Y_i = \alpha + \beta_1 X_i + \varepsilon_i$$

Residual for any candidate line (a, b):

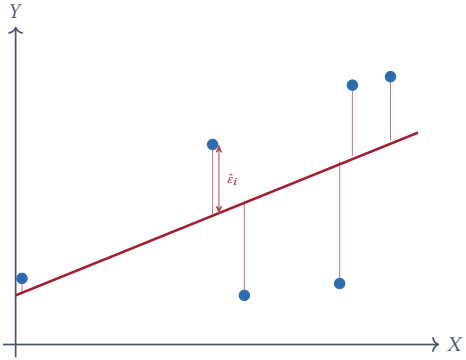
$$e_i(a, b) = Y_i - a - bX_i$$

OLS finds the minimizing pair:

$$(\hat{\alpha},\ \hat{\beta}_1) = \arg\min_{a,\,b} \sum_{i=1}^{n} e_i^2$$

The OLS residual: $\hat{\varepsilon}_i = Y_i - \hat{\alpha} - \hat{\beta}_1 X_i$ computed from data, unlike ε_i

The OLS problem: find the line that minimizes squared vertical gaps



Choose (a, b) to minimize:

$$\text{SSR}(a, b) = \sum_{i=1}^{n} \hat{\varepsilon}_i^2$$

where $\hat{\varepsilon}_i = Y_i - a - bX_i$

$$(\hat{\alpha}, \hat{\beta}_1) = \arg\min_{a,\, b} \ \text{SSR}(a, b)$$

FOC 1 (w.r.t. a): residuals sum to zero

Differentiate SSR with respect to a, set to zero:

$$\frac{\partial \text{SSR}}{\partial a} = -2 \sum_{i=1}^{n} (Y_i - \hat{\alpha} - \hat{\beta}_1 X_i) = 0 \quad \Longrightarrow \quad \sum_{i=1}^{n} \hat{\varepsilon}_i = 0$$

Divide by n:

$$\bar{Y} - \hat{\alpha} - \hat{\beta}_1 \bar{X} = 0$$

$$\boxed{\hat{\alpha} = \bar{Y} - \hat{\beta}_1 \bar{X} \quad \textbf{The OLS line passes through } (\bar{X}, \bar{Y})}$$

Fair: $\bar{X} = 3.85$, $\bar{Y} = 54.88$. Given $\hat{\beta}_1$, $\hat{\alpha}$ is immediate.

FOC 2 (w.r.t. b): residuals are uncorrelated with X

Differentiate SSR with respect to b, set to zero:

$$\frac{\partial \text{SSR}}{\partial b} = -2 \sum_{i=1}^{n} X_i (Y_i - \hat{\alpha} - \hat{\beta}_1 X_i) = 0 \quad \Longrightarrow \quad \sum_{i=1}^{n} X_i \hat{\varepsilon}_i = 0$$

Substitute $\hat{\alpha} = \bar{Y} - \hat{\beta}_1 \bar{X}$ and rearrange:

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}$$

$$\boxed{\hat{\beta}_1 = \frac{\hat{S}_{XY}}{\hat{S}_{XX}} = \frac{\widehat{\text{Cov}}(X, Y)}{\widehat{\text{Var}}(X)}}$$

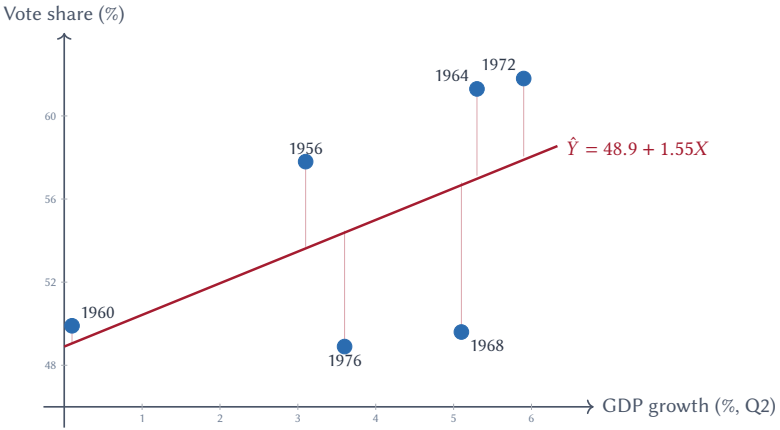
Two orthogonality conditions: $\sum \hat{\varepsilon}_i = 0$ and $\sum X_i \hat{\varepsilon}_i = 0$. Both reappear in matrix form.

Hand-computing $\hat{\beta}_1$ for the six Fair elections

Year	X_i	Y_i	$X_i - \bar{X}$	$Y_i - \bar{Y}$	$(X_i - \bar{X})(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$
1956	3.1	57.8	−0.75	+2.92	−2.19	0.56
1960	0.1	49.9	−3.75	−4.98	+18.68	14.06
1964	5.3	61.3	+1.45	+6.42	+9.31	2.10
1968	5.1	49.6	+1.25	−5.28	−6.60	1.56
1972	5.9	61.8	+2.05	+6.92	+14.18	4.20
1976	3.6	48.9	−0.25	−5.98	+1.50	0.06
Sum	23.1	329.3	0	0	**34.88**	**22.55**
Mean	3.85	54.88				

$$\hat{\beta}_1 = \frac{34.88}{22.55} \approx 1.55 \qquad \hat{\alpha} = 54.88 - 1.55 \times 3.85 \approx 48.9$$

The OLS line through Fair's six elections



1968: GDP growth 5.1% but vote share only 49.6%. Model predicts ≈57%. The residual is Vietnam and the Democratic convention.

Part II: Matrix Form

$$\hat{\beta} = (X'X)^{-1}X'Y$$

OLS imposes two orthogonality conditions on the residuals

Condition 1

$$\sum_{i=1}^{n} \hat{\varepsilon}_i = 0$$

Residuals average to zero

Line passes through $(\bar{X}, \bar{Y})$

Condition 2

$$\sum_{i=1}^{n} X_i \hat{\varepsilon}_i = 0$$

Residuals uncorrelated with X

X cannot predict the leftovers

OLS residuals are orthogonal to every variable in the model

X_i appears because the chain rule forces it out when we minimize SSR

Differentiate SSR with respect to b, set to zero:

$$\frac{\partial}{\partial b}\sum_i e_i^2 \;=\; \sum_i 2\left(Y_i - a - bX_i\right)\cdot\underbrace{\left(-X_i\right)}_{\substack{\text{chain}\\ \text{rule}}} \;=\; 0$$

Divide by -2:

$$\sum_i X_i\hat{\varepsilon}_i = 0 \qquad \Longleftrightarrow \qquad \widehat{\mathrm{Cov}}(X,\hat{\varepsilon}) = 0$$

$\hat{\beta}_1$ absorbs all the covariance between X and Y. Whatever remains in $\hat{\varepsilon}$ is orthogonal to X.

Both conditions hold in Fair's elections: verified with actual numbers

Year	X_i	Y_i	$\hat{\varepsilon}_i$	$X_i\hat{\varepsilon}_i$
1956	3.1	57.8	+4.1	+12.7
1960	0.1	49.9	+0.8	+0.1
1964	5.3	61.3	+4.2	+22.3
1968	5.1	49.6	−7.2	−36.7
1972	5.9	61.8	+3.7	+21.8
1976	3.6	48.9	−5.6	−20.2
Sum			**0.0**	**0.0**

Condition 1: $(+4.1) + (+0.8) + (+4.2) + (-7.2) + (+3.7) + (-5.6) = \mathbf{0.0}$

Condition 2: $3.1(4.1) + 0.1(0.8) + 5.3(4.2) + 5.1(-7.2) + 5.9(3.7) + 3.6(-5.6) = \mathbf{0.0}$

Each observation is a row in X; transposing makes each variable a row in X'

Design matrix X (6×2):

$$X = \begin{bmatrix} 1 & 3.1 \\ 1 & 0.1 \\ 1 & 5.3 \\ 1 & 5.1 \\ 1 & 5.9 \\ 1 & 3.6 \end{bmatrix} \begin{matrix} \leftarrow 1956 \\ \leftarrow 1960 \\ \leftarrow 1964 \\ \leftarrow 1968 \\ \leftarrow 1972 \\ \leftarrow 1976 \end{matrix}$$

Transpose X' (2×6):

$$X' = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 3.1 & 0.1 & 5.3 & 5.1 & 5.9 & 3.6 \end{bmatrix}$$

Row 1 $= [1, 1, \ldots, 1]$ intercept column of X

Row 2 $= [X_1, \ldots, X_6]$ GDP column of X

Row j of X' = column j of X = the multipliers from FOC j

$X'\hat{\varepsilon} = 0$: Fair's numbers confirm both conditions in one matrix product

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 3.1 & 0.1 & 5.3 & 5.1 & 5.9 & 3.6 \end{bmatrix}}_{X',\ 2\times 6} \underbrace{\begin{bmatrix} +4.1 \\ +0.8 \\ +4.2 \\ -7.2 \\ +3.7 \\ -5.6 \end{bmatrix}}_{\hat{\varepsilon},\ 6\times 1} = \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}$$

k regressors $\Rightarrow$ k rows in X' $\Rightarrow$ k orthogonality conditions at once. **That is why we need matrix notation.**

Preview: $(X'X)\hat{\beta} = X'Y$ — variation times slope equals co-movement.

With three regressors, scalar notation becomes unmanageable

Scalar: 6 equations, 4 unknowns

$$
\begin{aligned}
57.8 &= \beta_0 + \beta_1(3.1) + \beta_2(2.1) + \beta_3(1) + \varepsilon_1 \\
49.9 &= \beta_0 + \beta_1(0.1) + \beta_2(2.5) + \beta_3(0) + \varepsilon_2 \\
&\vdots
\end{aligned}
$$

Solve for $(\beta_0, \beta_1, \beta_2, \beta_3)$ simultaneously? Painful.

Matrix: one equation

$Y = X\beta + \varepsilon$, where

$$
Y = \begin{pmatrix} 57.8 \\ 49.9 \\ \vdots \\ 48.9 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 3.1 & 2.1 & 1 \\ 1 & 0.1 & 2.5 & 0 \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}
$$

$$
\boxed{Y : n \times 1 \quad X : n \times k \quad \beta : k \times 1}
$$

The Bridge: Moving from Scalar to Matrix

The problem with the scalar FOCs:

- You have to track $k + 1$ separate equations.
- Each equation requires calculating a different set of sums ($\sum X_{i1}\epsilon_i$, $\sum X_{i2}\epsilon_i$, etc.).

The Matrix Solution: We stack our variables into X and our outcomes into Y. The operation $X'X$ and $X'Y$ is effectively a **high-speed aggregator** that performs all those summations simultaneously.

Matrix algebra is just a "bulk" version of the summation operator $\sum$. It does all the bookkeeping for us in one click.

Visualizing the "Summing Machine"

When we write $X'X$, think of the rows of X' "sweeping" across the columns of X:

$$\underbrace{\begin{pmatrix} 1 & 1 & \dots \\ X_1 & X_2 & \dots \end{pmatrix}}_{X'} \underbrace{\begin{pmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \end{pmatrix}}_{X} = \begin{pmatrix} \sum 1 \cdot 1 & \sum 1 \cdot X_i \\ \sum X_i \cdot 1 & \sum X_i^2 \end{pmatrix} = \begin{pmatrix} n & \sum X_i \\ \sum X_i & \sum X_i^2 \end{pmatrix}$$

- The first row of X' (the 1s) creates the **sample size** (n) and the **total sum** ($\sum X$).
- The second row of X' (the data) creates the **sum of squares** ($\sum X^2$).

Why "Encode"? (The Information Perspective)

Why do we say these matrices "encode all the variation"?

- To find a line of best fit, you don't actually need to know if the 4th observation was $(x = 2, y = 5)$ or $(x = 5, y = 2)$.
- You only need the **Total Variation** $(X'X)$ and the **Total Co-variation** $(X'Y)$.

Once you calculate $X'X$ and $X'Y$, you have moved from N pieces of information (potentially millions of rows) down to just $(k + 1)^2$ numbers. This is "Data Compression" for Statistics.

$X'X$ and $X'Y$ encode all the variation we need

Bivariate Fair model (intercept + GDP growth only, 6 elections):

$$X'X = \begin{pmatrix} n & \sum X_i \\ \sum X_i & \sum X_i^2 \end{pmatrix} = \begin{pmatrix} 6 & 23.1 \\ 23.1 & 111.5 \end{pmatrix}$$

$(1,1) = n; \quad (1,2) = \sum X_i = 23.1$

$(2,2) = \sum X_i^2 = 111.5$

$$X'Y = \begin{pmatrix} \sum Y_i \\ \sum X_i Y_i \end{pmatrix} = \begin{pmatrix} 329.3 \\ 1302.7 \end{pmatrix}$$

$(1) = \sum Y_i = 329.3$

$(2) = \sum X_i Y_i = 1302.7$

$X'X$: variation in X $\qquad$ $X'Y$: covariation of X and Y

The normal equations lead directly to $\hat{\beta} = (X'X)^{-1}X'Y$

Minimize $(Y - X\beta)'(Y - X\beta)$. Differentiate, set to zero:

$$-2X'Y + 2X'X\hat{\beta} = 0 \quad \Longrightarrow \quad \underbrace{X'X\hat{\beta} = X'Y}_{\textbf{normal equations}}$$

Pre-multiply by $(X'X)^{-1}$**:**

$$\boxed{\hat{\beta} = (X'X)^{-1}X'Y}$$

The scalar FOCs from Part I are these two equations written out entry by entry. Matrix form is the same arithmetic, organized.

Step-by-step: $(X'X)^{-1}X'Y$ for the Fair elections

Step 1 — Invert $X'X$:

$$\det = 6(111.5) - 23.1^2 = 135.4$$

$$(X'X)^{-1} \approx \begin{pmatrix} 0.823 & -0.171 \\ -0.171 & 0.044 \end{pmatrix}$$

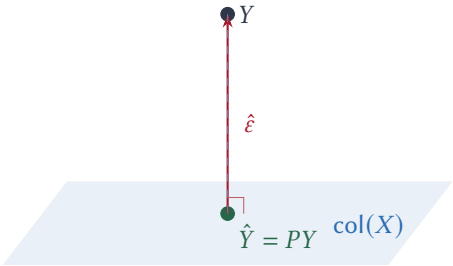
Step 2 — Multiply $(X'X)^{-1}X'Y$:

$$\hat{\beta}_0 = 0.823(329.3) - 0.171(1302.7) \approx 48.9$$

$$\hat{\beta}_1 = -0.171(329.3) + 0.044(1302.7) \approx 1.55$$

$(X'X)^{-1}X'Y$ gives the same answer as the scalar formula: $\hat{\beta}_1 \approx 1.55$, $\hat{\beta}_0 \approx 48.9$

OLS is a projection: $\hat{Y}$ is the shadow of Y onto col(X)



$$P = X(X'X)^{-1}X' \quad \text{(hat matrix)}$$

$$\hat{Y} = PY \qquad \hat{\varepsilon} = (I - P)Y$$

$$X'\hat{\varepsilon} = 0$$

Orthogonality is the projection condition

Part III: Model Fit

How well does the model explain the data?

$\hat{\beta}$ varies across samples — it has a sampling distribution

Decompose:

$$\hat{\beta} = (X'X)^{-1}X'Y = (X'X)^{-1}X'(X\beta + \varepsilon) = \beta + (X'X)^{-1}X'\varepsilon$$

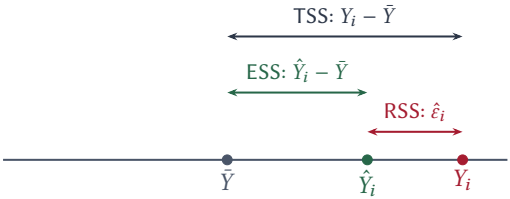
Rewrite as sample means:

$$\hat{\beta} - \beta = \underbrace{\left(\frac{X'X}{n}\right)^{-1}}_{\to Q^{-1}\ \text{(LLN)}} \cdot \underbrace{\frac{1}{n}\sum_i X_i\varepsilon_i}_{\to N(0,\Sigma)\ \text{(CLT)}}$$

$\hat{\beta} \approx N(\beta,\ Q^{-1}\Sigma Q^{-1}/n)$ — approximately normal because $\hat{\beta} - \beta$ is a sample mean

$\hat{\beta}$ is not normal because Y is normal. It is normal because $\hat{\beta} - \beta$ is linear in $\frac{1}{n}\sum_i X_i\varepsilon_i$. Slutsky delivers the rest.

Variance decomposes: total = explained + residual



$$\underbrace{(Y_i - \bar{Y})}_{\text{TSS}_i} = \underbrace{(\hat{Y}_i - \bar{Y})}_{\text{ESS}_i} + \underbrace{\hat{\varepsilon}_i}_{\text{RSS}_i}$$

$$\boxed{\text{TSS} = \text{ESS} + \text{RSS}}$$

Holds because $\hat{\varepsilon} \perp \hat{Y}$
(consequence of $X'\hat{\varepsilon} = 0$)

R^2 measures the fraction of variance the model explains

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

Fair model	Regressors	R^2	
GDP growth only	1	0.44	economy alone
GDP + inflation + incumbent	3	0.73	+29 pp from two regressors

$R^2 \in [0, 1]$. Zero: model explains nothing. One: perfect fit.

R^2 always rises when you add regressors — adjusted R^2 penalizes

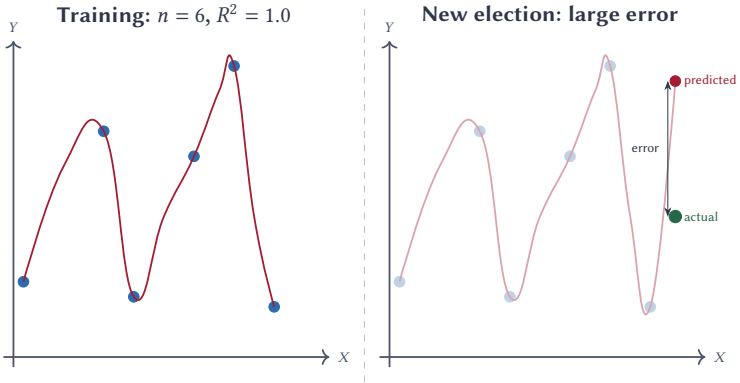
```r
# Full model: R^2 = 0.730
fit1 <- lm(vote ~ gdpg + inf + inc,
           data = fair)

# Add random noise
fair$noise <- rnorm(24)
fit2 <- lm(vote ~ gdpg + inf + inc
             + noise, data = fair)

summary(fit2)$r.squared       # 0.731
summary(fit2)$adj.r.squared   # 0.660
```

$$\bar{R}^2 = 1 - \frac{\text{RSS}/(n-k)}{\text{TSS}/(n-1)}$$

	R^2	$\bar{R}^2$
3 regressors	0.730	0.680
+ noise	0.731	0.660

Adding noise raises R^2; $\bar{R}^2$ falls

Perfect R^2 on training data can predict new data badly



R^2 measures fit on the data you used. It says nothing about new data.

This is why regularization (LASSO, Ridge) was invented. We return to it at the end of the course.

Part IV: Gauss–Markov

When is OLS the Best Linear Unbiased Estimator?

Four assumptions make OLS optimal

1. **Linearity:** $Y = X\beta + \varepsilon$ — linear in parameters
2. **Full rank:** $\text{rank}(X) = k$ — no perfect multicollinearity
3. **Strict exogeneity:** $\mathbb{E}[\varepsilon \mid X] = 0$ — mean-zero errors given X
4. **Spherical errors:** $\text{Var}(\varepsilon \mid X) = \sigma^2 I_n$ — homoskedastic, no autocorrelation

Under (1)–(4): OLS is **BLUE** — Best Linear Unbiased Estimator

Assumption 3 delivers unbiasedness: $\mathbb{E}[\hat{\beta} \mid X] = \beta$

$$\hat{\beta} = \beta + (X'X)^{-1}X'\varepsilon$$

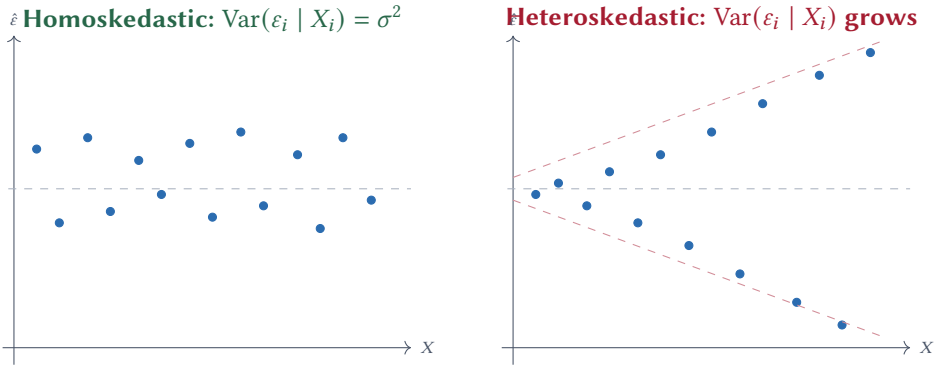
Take $\mathbb{E}[\,\cdot \mid X]$**:**

$$\mathbb{E}[\hat{\beta} \mid X] = \beta + (X'X)^{-1}X' \underbrace{\mathbb{E}[\varepsilon \mid X]}_{=0} = \beta$$

$\mathbb{E}[\hat{\beta} \mid X] = \beta$ — OLS is **unbiased** under strict exogeneity

Strict exogeneity rules out OVB, measurement error in X, and lagged-dependent-variable bias. Contemporaneous exogeneity ($\mathbb{E}[\varepsilon_i \mid X_i] = 0$) suffices for consistency.

Assumption 4: homoskedasticity vs. the fan



$\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I_n$: (a) homoskedasticity **and** (b) no autocorrelation

When A4 fails, OLS is still unbiased — but no longer efficient. Robust SEs fix inference. That is next lecture.

B-L-U-E: each letter, each assumption

	Word	Meaning	Needs	Breaks when
B	Best	Smallest variance	A4	heteroskedasticity
L	Linear	Linear in Y	A1	nonlinear CEF
U	Unbiased	$\mathbb{E}[\hat{\beta} \mid X] = \beta$	A3	OVB, endogeneity
E	Estimator	Function of data	A2	perfect collinearity

Gauss–Markov: OLS has the smallest variance among all linear unbiased estimators under A1–A4

OLS is the sample BLP — and it is optimal when the four conditions hold

$$\hat{\beta} = (X'X)^{-1}X'Y \quad \textbf{(plug-in estimator)}$$

Unbiased and BLUE under Gauss–Markov

Approximately normal because $\hat{\beta} - \beta$ is linear in $\frac{1}{n}\sum_i X_i \varepsilon_i$

Monday (12b):

Hat matrix, leverage, FWL in matrix form

Wednesday (13a):

SEs, t-tests, F-tests, robust SEs